

HAO QI

Ph.D. Student in Applied Mathematics (Statistics) · UNC Charlotte
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EDUCATION

University of North Carolina at Charlotte (UNC Charlotte) Ph.D. in Applied Mathematics (Statistics Concentration) · Advisor: Dr. Qingning Zhou	Charlotte, NC 08/2026 - Present
Rice University Master of Statistics · GPA: 3.96/4.00	Houston, TX 08/2024 - 12/2025
University of North Carolina Wilmington (UNCW) B.S. in Mathematics (Statistics Minor) · GPA: 3.94/4.00 · Dean's List	Wilmington, NC 08/2023 - 05/2024
Chongqing University of Arts and Sciences (CUAS) B.S. in Mathematics and Applied Mathematics · GPA: 91.1/100 · Rank: 1/95	Chongqing, China 09/2020 - 06/2024

PUBLICATIONS

- Xinjie Lan, Yunbo Mei, Leheng Zhou, **Hao Qi**, Shichen Tang, and Yunli Su. "Bayesian Time-aware Attention: Enhancing Transformers for Non-Stationary Time Series Forecasting." Manuscript submitted to AAAI, 2026.
- Travis J. Miles, Michael T. Guinn, Xin Tan, **Hao Qi**, Vicente Orozco-Sevilla, Marc R. Moon, et al. "Tissue perfusion pressure: A novel hemodynamic measure to assess risk of acute kidney injury after cardiac surgery." *J. Thorac. Cardiovasc. Surg.*, 171(2):455-462.e3, 2026. doi:10.1016/j.jtcvs.2025.07.009.
- Hao Qi** and Yuanshen Wang. "Prediction and Analysis of Stock Logarithmic Returns Based on ARMA-GARCH Model." *Proceedings of CECNet 2022*. doi:10.3233/FAIA220575.
- Hao Qi**. "Research on Forecasting for Different Types of Small Sample Time Series Data." *2022 6th Annual International Conference on Data Science and Business Analytics (ICDSBA)*. doi:10.1109/ICDSBA57203.2022.00035.

RESEARCH EXPERIENCE

Longitudinal Variational Autoencoder for Within-Subject Image Prediction, Rice University Supervised by Dr. Huixia Judy Wang (Rice University)	02/2026 - Present
- Developed a longitudinal variational autoencoder with a mixed-effects latent backbone and a first-order stochastic transition to model subject-specific temporal dynamics in image sequences. - Reformulated the task as prefix-conditioned missing-visit prediction, generating each visit from observed prior visits under a sequential latent-variable framework. - Derived an ELBO-based objective and evaluated one-step and recursive forecasting performance on longitudinal imaging data.	
A Novel Hemodynamic Measure to Assess Risk of Acute Kidney Injury After Cardiac Surgery, Rice & BCM	02/2025 - 07/2025
Supervised by Dr. Meng Li (Rice University)	
- Applied univariate and multivariable logistic regression to quantify the association between tissue perfusion pressure (TPP) and postoperative acute kidney injury (AKI), with analyses stratified by vasoactive inotrope score. - Identified a 38 mm Hg TPP threshold using regression-based risk estimation and demonstrated that TPP below this level remained an independent predictor after statistical adjustment. - Conducted prespecified sensitivity and subgroup analyses that yielded consistent effect estimates.	

WORK EXPERIENCE

Rice University · Research Assistant (with Dr. Meng Li)	01/2026 - 06/2026
- Developed dynamic machine-learning models to predict acute kidney injury after cardiac surgery using hemodynamics, laboratory measurements, medications, and clinical data. - Implemented rolling-window feature extraction and evaluated XGBoost, random forest, and logistic regression under strict patient-level data splits. - Built calibration-based evaluation metrics, mapped predicted creatinine to KDIGO AKI stages, and conducted external validation using the MIMIC dataset.	

AWARDS AND HONORS

China National Scholarship · Chongqing Outstanding College Graduate Student

SKILLS

Programming: Python, R, MATLAB, SAS | **Tools:** Word, Excel, PowerPoint, LaTeX | **Languages:** Mandarin (native), English (senior)